SMALL FRACTIONS WITH SMOOTH DENOMINATORS ARE THE SUM OF SEVEN UNIT FRACTIONS

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Extending the (in)famous Erdős-Straus conjecture, Schinzel [5] conjectured that any fraction $a/b \in (0,1)$ can be written as the sum of three unit fractions, as long as b is sufficiently large in terms of a. Although it is known that this holds for almost all b, even with three replaced by two (see e.g. [1] and [2]), it is currently not even known if a/b can be written as the sum of fewer than a unit fractions for all large b. Here we prove the following theorem, which gives a bounded sum of unit fractions for all sufficiently large smooth denominators b.[1]

Theorem. *For every positive integer a and every y there exists a $b_0 = b_0(a,y)$ such that all fractions a/b, whose denominator b is y-smooth and larger than b_0, can be written as the sum of at most 7 unit fractions.*[2]

Proof. Let a and y be given and let p be any prime. We then use [3, Theorem 1 and Table 1] which states that the set of primes

$$\{q : q \mid (p^t + 1) \text{ for some } t\}$$

has relative density at least $2/3 > 5/8$. This implies by [4, Theorem 1.1], that every sufficiently large odd integer can be written as the sum of three such primes q. In particular, we find that there exist primes q_1, q_2, q_3 and exponents s, t_1, t_2, t_3 (all dependent on p) such that

$$ap^s = \varepsilon + q_1 + q_2 + q_3 \qquad \text{and} \qquad p^{t_i} + 1 = d_i q_i,$$

for some $\varepsilon \in \{0,1\}$ and some positive integers d_i. With $b_0 := \prod_{p \le y} p^{\max_i t_i}$, we see that a y-smooth $b > b_0$ must be divisible by a prime power $p^r \ge p^{\max_i t_i}$, say $b = dp^r$. We then find

$$\begin{aligned}
\frac{a}{b} &= \frac{ap^s}{dp^{r+s}} \\
&= \frac{\varepsilon + q_1 + q_2 + q_3}{dp^{r+s}} \\
&= \frac{\varepsilon}{p^s} + \sum_{i=1}^{3} \frac{p^{t_i} + 1}{dd_i p^{r+s}} \\
&= \frac{\varepsilon}{p^s} + \sum_{i=1}^{3} \left(\frac{1}{dd_i p^{r+s}} + \frac{1}{dd_i p^{r+s-t_i}} \right),
\end{aligned}$$

as desired. □

[1] The author initially thought he had an idea to prove this result. ChatGPT then kindly pointed out to him that the idea does not work, before fixing it and giving a correct proof. The current note explaining the very short proof is human-generated.

[2] We remark that Takenouchi's theorem [6, Sections II–III] implies that one can take these unit fractions to be distinct as well, if one so desires.

References

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